A REMARK ON THE CONSTANT IN THE OPTIMAL-ORDER MIXING OF THE THORP SHUFFLE

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Abstract. The OpenAI Math Release preprint *Optimal-order mixing of the Thorp shuffle* proves that $q_d^{*(1600d)}$ converges to uniform on S_{2^d}; the best explicit constant elsewhere in the same family of preprints is $512d$. We show that the same conditional-flow method gives $\|q_d^{*\lfloor cd\rfloor} - U\|_{\mathrm{TV}} \to 0$ for every $c > 2/\log_2(8/7) + 2/\log_2(10/9) = 23.5394\ldots$, so $24d$ shuffles suffice for large d. Using only the paper's own operator-norm transfer (no companion results) the same argument gives every $c > 44.3006\ldots$ $(45d)$. The improvements are: an arbitrarily small concentration loss, a Kullback–Leibler chain rule in place of the sequential total-variation sum, a local look-back that reduces the start-up offset from $2d$ to $O(\log d)$, and unequal numbers of label blocks in different convolution factors. All numerical constants are certified with interval arithmetic. The result is asymptotic; no explicit threshold d_0 is given.

Attribution. The construction, the conditional-flow method and the transfer arguments are OpenAI's work (family 238 of `https://github.com/openai/math`, commit `08fcd618213aa43b0fcaff4526a8a48d66d02e55`): the main preprint [1] and the companion [2]. This note only re-does the bookkeeping of constants. We take the correctness of [1, 2] as given and refer to their numbered statements. Developed with AI assistance (Anthropic Claude, Opus 5.5) under [AUTHOR NAME]'s direction; independent adversarial reviews were run by separate AI sessions and are not human peer review.

1. Statement

Let $n = 2^d$, let q_d be the law of one Thorp shuffle as a permutation of positions and U the uniform law on S_n. Put

$$L(b) = \log_2 \frac{2b}{2b-1}, \qquad \kappa(b) = \frac{2}{L(b)}, \qquad e(b) = 1 - \frac{2}{b} \qquad (b \geq 3).$$

Theorem 1.1. *Let $b_1, \ldots, b_r \geq 3$ be integers and let $c > \sum_i \kappa(b_i)$.*

(H) *If $\sum_i e(b_i) > 1$, then $\|q_d^{*\lfloor cd\rfloor} - U\|_{\mathrm{TV}} \to 0$ as $d \to \infty$.*
(O) *If $\sum_i e(b_i) > 2$, the same conclusion holds, and its proof uses only the operator-norm transfer of [1] (no result of [2]).*

Corollary 1.2. *$\|q_d^{*\lfloor cd\rfloor} - U\|_{\mathrm{TV}} \to 0$ for every $c > c_H := \kappa(4) + \kappa(5) = 23.53941\ldots$ (route H), and, using only [1], for every $c > c_O := \kappa(3) + \kappa(4) + 2\kappa(5) = 44.30060\ldots$ (route O). In particular $t_{\mathrm{mix}}(d) \leq 24d$ for all large d. Within each route these block choices are optimal among integer block numbers; every admissible multiset costs more than $2\kappa(4) = 20.7636\ldots$ (route H) and $4\kappa(4) = 41.527\ldots$ (route O), and even the fractional relaxation over real b cannot go below 20.738 (route H) and 41.476 (route O); see Section 5.*

Comparison: [1, Thm. 1.1] has $1600d$; [3, Cor. 11.4] has $512d$; the lower bound is $2d - O(1)$ [1, Thm. 1.1]. Ratios: $1600/c_H = 67.97$, $512/c_H = 21.75$.

What is new, and what is not. The $512d$ bound of [3, Cor. 11.4] (label `shear:cor:512`) is already "first stage + the companion's two-factor Hilbert–Schmidt transfer" ($2 \times 256d$, eight equal

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blocks). Hence the factor 4 that the two-factor transfer contributes in Table 1 is *not* a contribution of this note. What this note adds is (i) three improvements of the first stage: concentration loss $\theta \to 0$ (Lemma 2.2), the Kullback–Leibler chain rule (Lemma 2.4) and the local look-back (Lemmas 2.1–2.3); and (ii) different block numbers $\{4, 5\}$ in the two factors. Measured against the best constant in the family, these give $512d \to 23.54d$, a factor 21.75. Route O is the version that uses [1] alone.

2. First stage: averaged ordered marginals

We use the notation of [1, Sections 2–3]: the cyclic-coordinate frame, the exposed list $c_1, \dots, c_\ell$, the free set B_s with $|B_s| = m = n - \ell$, the centered flow $w_s = p_s - \frac{1}{m}\mathbf{1}_{B_s}$, $E_s = \mathbb{E}\|w_s\|^2$ and $b_s = \mathbb{E}\sum_x w_s(x)^2 \mathbf{1}\{x^{(i_s)} \notin B_s\}$; expectations include a uniformly random ordered list, independent of the shuffle.

Lemma 2.1 (Energy identity including the initial term)**.** *For all $t \geq 0$, $E_t = I_t + \sum_{j=1}^{\min(t,d)} 2^{-j} b_{t-j}$, where $I_t = 0$ for $t \geq d$ and $I_t = \mathbb{E}\|P_t w_0\|^2 \leq 2^{-t} + 1/m$ for $t < d$; here $P_t = A_{i_{t-1}} \cdots A_{i_0}$.*

Proof. For $t \geq d$ this is [1, Lemma 3.2]. For $t < d$ expand [1, (3.1)] down to time 0: $w_t = P_t w_0 + \sum_{j=1}^{t} A_{i_{t-1}} \cdots A_{i_{t-j+1}} \xi_{t-j}$. The cross terms vanish exactly as in [1, Lemma 3.2] ($P_t w_0$ is $\mathcal{F}_0$-measurable and each noise has conditional mean zero), and the j-th noise contributes $2^{-j} b_{t-j}$ because the $j-1$ later averages use distinct coordinates. Since P_t is an orthogonal projection and $w_0 = \delta_c - \frac{1}{m}\mathbf{1}_{B_0}$, $\|P_t w_0\|^2 = \langle w_0, P_t w_0\rangle \leq \langle \delta_c, P_t \delta_c\rangle + \frac{1}{m^2}\langle \mathbf{1}_{B_0}, P_t \mathbf{1}_{B_0}\rangle \leq 2^{-t} + \frac{1}{m}$ (the middle cross term is ≤ 0). □

Lemma 2.2 (Contraction with local look-back)**.** *Fix $\theta \in (0, 1)$ and put $\beta = (1-\theta)m/n$. For every $s \geq 0$,*

$$b_s \leq (1-\beta)E_s + \beta\eta_s, \qquad \eta_s = \frac{n}{2^\sigma}\exp\Big(-2^{\sigma+1}\theta^2\frac{m^2}{n^2}\Big), \quad \sigma = \min(s, d-1).$$

Proof. Let $a = s - \sigma$ (so $a = 0$ if $s < d-1$, and $a = s - d + 1$ otherwise, as in [1, Section 4]). The updates at times $a, \dots, s-1$ use the σ distinct directions $i_a, \dots, i_{s-1}$, none equal to i_s, so they do not cross the halves H_0, H_1 of direction i_s, and conditional on $\mathcal{F}_a$ the two halves evolve independently. A free position at time a is moved to a uniform point of the σ-dimensional subcube $Q(y)$ spanned by these directions; hence $\Pr\{y \in B_s \mid \mathcal{F}_a\} = |B_a \cap Q(y)|/2^\sigma$, where $Q(y)$ lies in the half opposite to the partner x of y. The proof of [1, (4.1)] applies verbatim and gives $\mathbb{E}[\sum_x w_s(x)^2 \mathbf{1}\{x^{(i_s)} \in B_s\} \mid \mathcal{F}_a] \geq \beta\,\mathbb{E}[\|w_s\|^2 \mid \mathcal{F}_a]$ on the $\mathcal{F}_a$-measurable event G that every such subcube has $|B_a \cap Q| \geq (1-\theta)2^\sigma m/n$. Since $\|w_s\|^2 \leq 1$, $b_s \leq E_s - \beta(E_s - \Pr(G^c))$. For a random list B_a is a uniform m-subset [1, Section 4], so each count is hypergeometric with 2^σ draws; Hoeffding's bound for sampling without replacement [4, Thm. 4] gives $\exp(-2 \cdot 2^\sigma \theta^2 m^2/n^2)$ per subcube, and there are $n/2^\sigma$ subcubes. □

Lemma 2.3 (Closed form)**.** *Let $\gamma' \in (1 - \beta/2, 1)$, $\rho = (1-\beta)/(2\gamma' - 1) < 1$, let $s_0 \leq d$ and $\eta^* = \max_{s \geq s_0} \eta_s$. Then for all $t \geq 0$*

$$E_t \leq \frac{2}{1-\rho}\gamma'^{\,t-s_0} + \frac{\gamma'^{\,t-d}}{m(1-\rho)} + \eta^*.$$

Proof. Call the right side U_t. For $t < s_0$, $U_t \geq 2 > 1 \geq E_t$. Let $t \geq s_0$ and assume $E_s \leq U_s$ for $s < t$. In Lemma 2.1 use $b_s \leq E_s \leq 1$ for $s < s_0$ and Lemma 2.2 for $s \geq s_0$. The terms with $t - j < s_0$ contribute at most $\sum_{j > t-s_0} 2^{-j} \leq 2^{-(t-s_0)} \leq \gamma'^{\,t-s_0}$, and $I_t \leq 2^{-t} + \frac{1}{m} \leq \gamma'^{\,t-s_0} + \frac{1}{m}\gamma'^{\,t-d}$ for $t < d$. The remaining terms contribute at most $\sum_j 2^{-j}\big[(1-\beta)U_{t-j} + \beta\eta^*\big] \leq \rho\Big(\frac{2\gamma'^{\,t-s_0}}{1-\rho} + \frac{\gamma'^{\,t-d}}{m(1-\rho)}\Big) + (1-\beta)\eta^* + \beta\eta^*$, using $(1-\beta)\sum_{j \geq 1}(2\gamma')^{-j} = \rho$. Adding up, $E_t \leq U_t$ because $2 + \frac{2\rho}{1-\rho} = \frac{2}{1-\rho}$ and $1 + \frac{\rho}{1-\rho} = \frac{1}{1-\rho}$. □

Lemma 2.4 (KL chain rule)**.** *For a random tuple $(Z_1,\dots,Z_k)$ of distinct positions and σ-fields $\mathcal{G}_j \supseteq \sigma(Z_1,\dots,Z_{j-1})$ with $p_j = \mathcal{L}(Z_j \mid \mathcal{G}_j)$ supported on the m_j free positions, $\|\mathcal{L}(Z) - U_{\mathrm{inj}}\|_{\mathrm{TV}} \le \big(\frac{1}{2}\sum_j m_j \mathbb{E}\|p_j - U_{m_j}\|_2^2\big)^{1/2}$.*

Proof. The chain rule gives $D(\mathcal{L}(Z)\|U_{\mathrm{inj}}) = \sum_j \mathbb{E} D(\mathcal{L}(Z_j \mid Z_{<j})\|U_{V\setminus Z_{<j}})$. Since the reference law is $Z_{<j}$-measurable, convexity of D in its first argument lets us refine the conditioning to $\mathcal{G}_j$. Then $D(p\|U_m) \le \log(1+\chi^2) \le \chi^2 = m\|p - U_m\|_2^2$, and Pinsker's inequality finishes the proof. □

Proposition 2.5. *Fix an integer $b \ge 3$ and $c_b > \kappa(b)$, and put $T = \lfloor c_b d \rfloor$. For lists of length $k \le n - \lfloor n/b \rfloor$ chosen uniformly and independently of the shuffle, $\mathbb{E}_C\|\mathcal{L}(\Pi_T(c_1),\dots,\Pi_T(c_k) \mid C) - U_{\mathrm{inj}}\|_{\mathrm{TV}} \to 0$, uniformly in such k.*

Proof. Every exposure stage has $m_j \ge n-k+1 > n/b$. Apply Lemma 2.4 with $\mathcal{G}_j$ generated by the list and the exposed paths (as in [1, Prop. 4.2]) and average over the list by Jensen: the left side is at most $n\,(\max_j E_T^{(j)}/2)^{1/2}$. Choose θ small and γ' close to $1-(1-\theta)/(2b)$ so that $c_b \log_2(1/\gamma') > 2$; then $\beta \ge (1-\theta)/b$ and $\rho < 1$ for all j. Take $s_0 = \lceil \log_2(Kd) \rceil$ with $K > \frac{3}{2}b^2 \ln 2/\theta^2$. Then $\eta^* \le n\exp(-2Kd\theta^2/b^2) = o(n^{-2})$, $n^2\gamma'^{T-s_0} \to 0$ since $s_0 = O(\log d)$, and $n^2\gamma'^{T-d}/m \le bn\gamma'^{T-d} \to 0$ because $(c_b-1)\log_2(1/\gamma') > 2 - \log_2(1/\gamma') > 1$. Lemma 2.3 gives $n^2 \max_j E_T^{(j)} \to 0$. □

3. Second stage: transfers

Fix b, sizes $\lfloor n/b \rfloor$ or $\lceil n/b \rceil$ for b label blocks $M_1,\dots,M_b$ and $H_a = \mathrm{Sym}(M_a)$. For $\mu = q_d^{*T}$ with T as in Proposition 2.5, averaging over a uniformly random ordered partition with these sizes (each complement is a uniform list of length $\le n - \lfloor n/b \rfloor$) gives a partition with $\delta_n := \sum_a \|\mu|_{V\setminus M_a} - U_{\mathrm{inj}}\|_{\mathrm{TV}} \to 0$. The trimming of [1, Section 6, (6.2)] then gives $\nu = \mu(\cdot \mid E)$ with $\|\nu - \mu\|_{\mathrm{TV}} = 1 - M \le 2\delta_n \to 0$ and $\nu(gH_a) \le B|H_a|/|G|$, $B = 2/M \le 4$, for all a. For $\sigma > 0$ let $C_\sigma = \sup_{s\ge 1} \sum_{\lambda\vdash s} (f^\lambda)^{-\sigma}$.

Lemma 3.1. *For every $\sigma > 0$, $C_\sigma < \infty$ and $\sum_{\lambda\vdash s,\ \lambda\neq(s),(1^s)} (f^\lambda)^{-\sigma} \to 0$.*

Proof. This is a special case of [5, Thm. 2.6]. The elementary proof of [1, Lemma 5.1] also works: its three cases give $f^\lambda \ge 3^u$, $f^\lambda \ge 2^{\lfloor s/16 \rfloor}$ and $f^\lambda \ge (4/e)^s$, and $\sum_u 2p(u)3^{-\sigma u} < \infty$, $e^{3\sqrt{s}}2^{-\sigma\lfloor s/16\rfloor} \to 0$, $e^{3\sqrt{s}}(4/e)^{-\sigma s} \to 0$ for every fixed $\sigma > 0$. □

Proposition 3.2. *For every irreducible ρ of S_n of degree D and every $\sigma > 0$:*

(O) $\|\widehat{\nu}(\rho)\|_{\mathrm{op}}^2 \le bBC_\sigma D^{-1+(2+\sigma)/b}$ [1, Prop. 6.1] *with $D^{1/8}, C_1$ replaced by $D^{1/b}, C_\sigma$ (this form is also stated in* [2, Thm. 1.1, (1.3)]*);*

(H) $\|\widehat{\nu}(\rho)\|_{\mathrm{HS}}^2 \le bBC_\sigma D^{-1+(2+\sigma)/b}$ [2, Thm. 1.1, (1.2)].

Proof. (O): in the proof of [1, Prop. 6.1], every tensor type has a factor of degree $\le D^{1/b}$, block sizes play no role, and $\dim W_a \le \sum_{\dim\pi \le D^{1/b}} (\dim\pi)^{2+\sigma}(\dim\pi)^{-\sigma} \le C_\sigma D^{(2+\sigma)/b}$. (H) is stated in [2] for disjoint nonempty blocks of arbitrary sizes and every $u > 0$; for the reader's convenience: if e_τ is the central idempotent of $\tau \in \widehat{H}_a$ and $P_\tau = \rho(e_\tau)$, Plancherel and $\|(\nu|_C) * e_\tau\|_2^2 = \langle \nu|_C, (\nu|_C) * e_\tau\rangle \le \nu(C)^2(\dim\tau)^2/|H_a|$ on each coset C give $D\|\widehat{\nu}(\rho)P_\tau\|_{\mathrm{HS}}^2 \le B(\dim\tau)^2$; summing over the orthogonal decomposition $V_\rho = \bigoplus_a E_a$ of [1, Prop. 6.1] yields (H). □

4. Proof of Theorem 1.1

Split $\lfloor cd \rfloor \ge \sum_i \lfloor c_i d \rfloor$ with $c_i > \kappa(b_i)$; extra steps do not increase the distance to uniform.

Frames. [1, Section 2] returns to the physical frame only after blocks whose length is a multiple of d, and $\lfloor c_i d \rfloor$ is in general not a multiple of d. This is harmless. With R the cyclic rotation of coordinates and $Y_t = R^t X_t$ [1, Section 2], the physical law of one block of length T is $\mu(g) = \mu^{\mathrm{cyc}}(R^{-T}g)$, i.e. the cyclic-frame law left-multiplied by the fixed position permutation R^T. Left

multiplication by a fixed $h \in S_n$ maps the marginal of any label list through the bijection h of positions, which fixes U_{inj}, so every marginal TV distance in Proposition 2.5 is unchanged; and it maps each coset gH_a to the coset hgH_a of the same label-block subgroup, so the coset masses $\max_g \nu(gH_a)$ (hence the trimming of Section 3 above and Proposition 3.2) are unchanged. We therefore apply Sections 2–3 above to the physical laws $\mu_i = q_d^{*\lfloor c_i d \rfloor}$ directly, and the convolution $\mu_1 * \cdots * \mu_r$ is the physical law $q_d^{*\sum_i \lfloor c_i d \rfloor}$. Let $\mu_i = q_d^{*\lfloor c_i d \rfloor}$ and ν_i its trimmed law (Section 3, with $b = b_i$). Choose $\sigma > 0$ so small that $\epsilon := \sum_i (1 - (2+\sigma)/b_i) - \tau > 0$, where $\tau = 1$ in case (H) and $\tau = 2$ in case (O). By the Plancherel bound [1, (7.1)] and $\|XY\|_{\mathrm{HS}} \le \|X\|_{\mathrm{HS}} \|Y\|_{\mathrm{op}}$, for $\rho \neq \mathbf{1}, \mathrm{sgn}$:

$$D \Big\| \prod_i \widehat{\nu}_i(\rho) \Big\|_{\mathrm{HS}}^2 \le \begin{cases} D \, \|\widehat{\nu}_1\|_{\mathrm{HS}}^2 \prod_{i \ge 2} \|\widehat{\nu}_i\|_{\mathrm{op}}^2 \le A \, D^{-\epsilon} & \text{(H)}, \\ D^2 \prod_i \|\widehat{\nu}_i\|_{\mathrm{op}}^2 \le A \, D^{-\epsilon} & \text{(O)}, \end{cases} \qquad A = \prod_i 4 b_i C_\sigma,$$

and $\sum_{\rho \neq \mathbf{1}, \mathrm{sgn}} D_\rho^{-\epsilon} \to 0$ by Lemma 3.1. The sign coefficient of each ν_i is at most $(1 - M_i)/M_i \to 0$ [1, Section 7]. Hence $\|\nu_1 * \cdots * \nu_r - U\|_{\mathrm{TV}} \to 0$, and $\|\mu_1 * \cdots * \mu_r - U\|_{\mathrm{TV}} \le \sum_i (1 - M_i) + \|\nu_1 * \cdots * \nu_r - U\|_{\mathrm{TV}} \to 0$. Translation invariance gives the same from every initial deck. □

Remark 4.1. In case (O) all factors are bounded through $\|B\|_{\mathrm{HS}} \le \sqrt{D} \|B\|_{\mathrm{op}}$, which costs one full power of D; this is why the main paper needs many factors. The companion's Hilbert–Schmidt bound removes that loss for one factor and reduces the requirement from $\sum e(b_i) > 2$ to $\sum e(b_i) > 1$.

5. Constants, certificate and loss ledger

Corollary 1.2 is a finite optimisation. The script `scripts/constants_certificate.py` enumerates all multisets with $\sum e(b_i)$ above the threshold (exact rationals) and evaluates $\sum \kappa(b_i)$ in `mpmath.iv` interval arithmetic. Completeness uses $L(b) \le 1/((2b-1)\ln 2)$, i.e. $\kappa(b) \ge 2(2b-1)\ln 2$. The certified optima are $(4, 5)$ with $c_H \in [23.539413097, 23.539413098]$ (runner-up $(3, 3, 4)$, 25.589) and $(3, 4, 5, 5)$ with $c_O \in [44.300608089, 44.300608090]$ (runner-up $(4, 4, 4, 5)$, 44.30299). The slack conditions are exact: $(4, 5)$ works for every $\sigma < 2/9$ and $(3, 4, 5, 5)$ for every $\sigma < 2/59$. Since $\sum \kappa(b_i) = \sum \frac{\kappa(b_i)}{e(b_i)} e(b_i)$, every admissible multiset costs more than $\tau \cdot \min_{b \ge 3 \text{ integer}} \kappa(b)/e(b) = \tau \cdot 2\kappa(4) = 20.7636\ldots\tau$. Over real $b \ge 3$ the fractional relaxation has infimum $\min_b \kappa(b)/e(b) = 20.738\ldots$ (certified in $[20.73813, 20.73818]$, minimiser $b^* \approx 3.8647$). These are strict lower bounds for the present first-stage rates, not attained values: with these ingredients route H cannot go below 20.738 and route O not below 41.476.

cumulative change (route H)	constant	step
paper as written: $8 \times 200d$	1600	
paper's own inequalities, minimal $T = (2 + 3/\log_2 \frac{32}{31})d$ per factor	539.98	2.96
companion's two-factor transfer, $b = 8$ (already in [3, Cor. 11.4])	134.99	4.00
concentration $\theta \to 0$: $\beta = 1/16 \to 1/8$	68.44	1.97
KL chain rule (need $n^2 E_T \to 0$, not $n^3 E_T \to 0$)	46.96	1.46
local look-back: offset $2d \to O(\log d)$	42.96	1.09
block numbers $\{4, 5\}$ instead of $\{8, 8\}$	23.54	1.82

Table 1. Where the constant goes (all values evaluated in interval arithmetic by `constants_certificate.py`). The step ×4 is the family's own (it is how $512d$ is obtained in [3]); the last four rows are this note's contribution. Route O (paper-only): 1600 → 337.5 (five factors suffice for the paper's Lemma 5.1) → 171.1 ($\theta \to 0$) → 117.4 (KL) → 107.4 (look-back) → 64.4 (any $\sigma > 0$, three factors with $b = 8$) → 44.30 (blocks $\{3, 4, 5, 5\}$).

6. Scope

Not claimed: an explicit d_0 (the constants C_σ for small σ are finite but not computed); any statement about the true leading constant (exact values $t_{\mathrm{mix}}(2) = 3$, $t_{\mathrm{mix}}(3) = 5$ are only unit tests); a Lean formalisation. Simulations of the energy recurrence (`energy_lemma_check.py`, `energy_mc.py`) only illustrate Lemmas 2.1–2.3.

References

[1] OpenAI, *Optimal-order mixing of the Thorp shuffle*, OpenAI Math Release preprint, September 26, 2026, https://github.com/openai/math/tree/main/preprints/Optimal-order-mixing-of-the-Thorp-shuffle-September-26-2026.

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